

TECHNICAL OVERVIEW

United4Change Client Tech Stack

A transparent, blockchain-backed donation platform: a Next.js web client that talks to a REST backend and to smart contracts on Polygon.

PROJECT	VERSION	PREPARED	SOURCE
u4c-client	0.1.0	28 September 2026	package.json & source code

1. At a glance

The client is a server-rendered React application (Next.js App Router, TypeScript) styled with Tailwind CSS. Data comes from a separate REST API; donations are pledged on-chain through Privy-managed wallets on Polygon. It is hosted on Vercel and ships as an installable PWA.

LAYER	TECHNOLOGY	VERSION	ROLE
Framework	Next.js (App Router)	^16.0.10	Routing, rendering, image optimisation, redirects, metadata. Dev and build both run with <code>--webpack</code> .
UI library	React	19.2.1	Component model; most pages are client components (<code>"use client"</code>).
Language	TypeScript	^5	Strict mode, path alias <code>@/*</code> to <code>src/*</code> .
Styling	Tailwind CSS	^4	Utility classes via <code>@tailwindcss/postcss</code> , plus shared classes in <code>src/style/globals.css</code> .
State & data	Redux Toolkit, RTK Query, react-redux, redux-persist	2.8 / 9.2 / 6.0	Global user state, API layer, and local persistence of the session.
HTTP	Axios	^1.11	Custom RTK Query base query that talks to the backend.
Auth & wallets	Privy	^3.21	Login, embedded and external wallets, smart wallets.
Blockchain	ethers, viem	^6.15 / ^2.47	Contract calls, chain configuration and token transfers on Polygon.
Hosting	Vercel	-	Deploys the Next.js app (build, serverless rendering and the CDN).

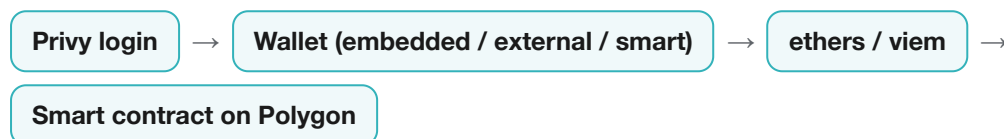
• 2. Application & data flow

Requests to the backend



Endpoints are grouped in `src/redux/api` into `public`, `private` and `ngo` files, all registered on one API slice. Each endpoint declares whether it needs a token (`requireToken`) and whether the body is multipart. The base URL is chosen by environment: a test backend in Development, the production backend otherwise.

Donations and wallets



Users sign in through Privy. Depending on the wallet type in the user state (`smart-wallet` or an external wallet) the app either uses a Privy smart-wallet client or the connected wallet to pledge USDC/USDT to a campaign contract. The network is switched by one setting: `Test-Net` uses Polygon Amoy (chain 80002), anything else uses Polygon mainnet (chain 137). The contract interface lives in `src/Wallet/wallet.abi.ts`.

Client state

Redux holds the signed-in user, the wallet type and the API cache. State is persisted to `localStorage` with `redux-persist` (versioned, with migrations); the API cache is excluded from persistence, and a no-op storage is used during server rendering.

• 3. Libraries by purpose

Interface & layout

`react-icons` (icons), `Swiper 12` (carousels), `react-select` (dropdowns), `react-toastify` (notifications).

Forms & media

`react-easy-crop` (image cropping for campaign banners), `react-international-phone`, `react-country-flag`, `world-countries`.

Charts & progress

`Chart.js 4` with `chartjs-plugin-datalabels` (used directly), `react-circular-progressbar`, `qrcode.react`.

Maps

`Leaflet`, `react-leaflet` and `leaflet-geosearch`. Tiles come from `OpenStreetMap`; addresses are geocoded to show the campaign location.

Motion

`GSAP` (animations) and `Lenis` (smooth scrolling).

Auth helpers

`jwt-decode` reads token claims in the API layer.

PWA

@ducanh2912/next-pwa generates the service worker (Workbox). It is switched off in development and caches front-end navigation aggressively in production.

SEO

`src/lib/seo.ts` centralises the site identity; `sitemap.ts`, `robots.ts` and `opengraph-image.tsx` generate the metadata. All traffic on `www` is permanently redirected to the apex domain.

4. Project structure

PATH	CONTENTS
<code>src/app</code>	Routes: home, <code>explore</code> , <code>campaign</code> , <code>ngo</code> , <code>initialize-campaign</code> , <code>in-app-donation</code> , sign-in / sign-up / password and e-mail verification pages, information pages (FAQ, how it works, contact) and policy pages (privacy, terms, AML/CTF, treasury).
<code>src/app/dashboard</code>	Signed-in area: campaigns (create, edit, overview, milestones, updates, gallery, comments), profile, settings, transactions and treasury.
<code>src/components</code>	Shared UI: campaign card, filters, pagination, maps, charts, and <code>dashboard/</code> components for NGO and donor views.
<code>src/Wallet</code>	Wallet connection, contract interaction, smart-contract ABI and the Privy provider setup (<code>privy/</code>).
<code>src/redux</code>	Store, persisted user slice and the API endpoints.
<code>src/lib</code> , <code>src/style</code>	SEO helpers and global CSS.

5. Environment configuration

Configuration is read from `.env.local`. Only the variable names are listed here; values are not included in this document.

VARIABLE	CONTROLS
<code>NEXT_PUBLIC_ENVIRONMENT</code>	Development or Production: selects which API base URL is used.
<code>NEXT_PUBLIC_DEV_API_LINK</code> / <code>NEXT_PUBLIC_PROD_API_LINK</code>	Backend base URLs for each environment.
<code>NEXT_PUBLIC_NETWORK</code>	<code>Test-Net</code> for Polygon Amoy, otherwise Polygon mainnet.
<code>NEXT_PUBLIC_PROJECT_ID</code>	Privy application ID.
<code>NEXT_PUBLIC_ALCHEMY_MAINNET_KEY</code> / <code>..._TESTNET_KEY</code>	Alchemy RPC access for each network.
<code>NEXT_PUBLIC_CONTRACT_ADDRESS</code> , <code>..._MAINNET</code> , <code>..._TESTNET</code>	Campaign contract address.

`NEXT_PUBLIC_USDC_*` / `NEXT_PUBLIC_USDT_*`

Token contract addresses, mainnet and testnet.

`NEXT_PUBLIC_URL` , `NEXT_PUBLIC_SITE_URL`

Site URL settings.

`NEXT_PUBLIC_SITE_URL` is optional and defaults to the production domain.

Security note. Every `NEXT_PUBLIC_` variable is bundled into the browser code, so the Alchemy keys are visible to anyone. Restrict them by allowed domain in the Alchemy dashboard.

• 6. Running the project

COMMAND	WHAT IT DOES
<code>npm run dev</code>	Development server (webpack), PWA disabled.
<code>npm run build</code>	Production build (webpack).
<code>npm run start</code>	Serve the production build.

Next.js 16 requires Node.js 20.9 or newer.

• 7. Observations & suggestions

Declared but not imported. A search of `src` found no imports of these packages: `@alchemy/aa-core` , `permissionless` , `ox` , `@solana/web3.js` , `@solana/spl-token` , `lucide-react` and `react-chartjs-2` . If they are not needed, removing them shrinks installs and lockfile noise. Note that the search covers static imports only, so please confirm before removing.

- **Lint script.** `npm run lint` is defined, but ESLint is not listed as a dependency, so the script may not run as expected.
- **Backend.** The backend is a separate service and is not part of this repository. Its error and filter conventions (for example `errors[0].detail` and `categories__name`) look like a Django REST Framework API, but this is an inference from the client code, not confirmed.
- **Netlify config.** The repository contains a `netlify.toml` (a 4 GB Node memory setting). Since the project is hosted on Vercel, this file has no effect there and can be removed if it is no longer needed.
- **Versions.** Versions above are the ranges declared in `package.json` ; the exact installed versions are pinned in `package-lock.json` .